

Smaran Vallabhaneni

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EDUCATION

Olin College of EngineeringNeedham, MA

Bachelor of Science, Engineering in Computing (GPA: 4.0)Aug. 2025 – May 2029

– **Relevant Courses:** Modeling & Simulation, Linear Algebra, Data Science, Multi-Variable Calculus, Software Design (Python), Kernel Development (C), Cryptography, Algorithms and Data Structures, Machine Learning, Products & Markets, Introduction to Sensors, Instrumentation & Measurement

SKILLS

Programming Languages: Java, C/C++, Python, pandas, HTML, CSS, JavaScript, Dart, L^AT_EX, MySQL, NoSQL, C#, R, PBASIC, Bash, Zsh, PowerShell, Swift, MATLAB

Operating Systems / SBCs: Windows, Unix, macOS, Linux (Raspberry Pi OS, Debian, Fedora, and Arch based), Raspberry Pi, Arduino

Tools: Blender, Fusion360, AutoCAD, Autodesk, SolidWorks, Unity, Figma, Adobe Creative Suite, Excel, Git, IntelliJ, Eclipse IDE, VSCode, Xcode, Overleaf, Node.js, GPL Cver, Simulink, MATLAB, Tableau Desktop, Docker, MissionPlanner, ArduPilot, GitHub Copilot, Google Gemini, Claude Code, LM Studio

Frameworks: LLMs, RAG Systems, Django, PyTorch, NumPy, Tkinter, Scikit-Learn, Kivy, Flutter, SwiftUI, React, AngularJS, .NET, Ollama, System Design, Metal API, ROS/MAVROS, OpenCV, MediaPipe

Other: UI/UX, CI/CD Pipelines, ESP32, Computer Vision, Internet of Things, AWS, GCP, Azure, Dev Ops, HuggingFace, NLP, Databases, Database Management, Statistical Analysis, Version Control, Embedded Systems, Algorithm Development, Data Structures, Object-Oriented Programming, API Integration, Testing and Debugging

Soft Skills: Communication, Problem Solving, Team Collaboration, Leadership, Adaptability, Time Management, Attention to Detail

PROJECTS

CLOS (Command Line Operating System) | C++, Java, HTML, CSS, BashApr. 2023 – Present

– Building a cross-platform centralized terminal and graphical math tool with equation solving, graphing, symbolic calculus, and trigonometry.

– Engineered a custom database engine, AES encryption module, symbolic math parser, and a 3D rendering library from scratch in Java; achieved consistent 60 FPS graphic rendering and executed complex multi-variable calculus computations under 50 ms.

ClassicTunes | Swift, SwiftUI, Metal APIOct. 2025 – Jul. 2026

– Developed an open-source modern remake of iTunes 7–10 for Apple Silicon macOS, faithfully recreating CoverFlow, MiniPlayer, the iTunes Store, iPod Syncing, and Smart/Genius Playlists.

– Leveraged Apple’s Metal API to build a custom GPU-accelerated CoverFlow engine, matching the performance of the native Apple Music client with a fluid 120 FPS render rate across 150+ albums on Apple Silicon.

Whoa-Scope | Python, KivyFeb. 2026 – Apr. 2026

– Co-developed an open-source oscilloscope fork to resolve compatibility issues and desktop UX flaws, gaining organic adoption from 50% of the class (40 students) and faculty recognition.

– Profiled and refactored memory allocation, slashing RAM consumption by 200+ MB to maximize stability and prevent lag on lab computers.

Specter | Python, Linux, LLMsMay 2026

– Created a local-first, privacy oriented AI screen agent for Fedora Wayland to perform tasks efficiently with simple prompts.

– Locally ran with a response time of 3 seconds per action, with extreme fidelity and understanding of on-screen content on a capable GPU.

3D Gesture Tracking | Blender, Python, OpenCVApr. 2026

– Engineered a real-time computer vision add-on published on Blender Addons, enabling mouse-free viewport manipulation (pan, zoom, rotate) via hand gesture tracking.

OSM2Mesh | Python, OpenStreetMapJul. 2026

– Built an easy-to-use map to 3D model city generator that produces high-quality, game-ready city models with textures within 20 seconds.

– Designed for all standard 3D and game engines with extensive customizability and an easy-to-use cross-platform UI.

EXPERIENCE

Olin College AERO Project TeamNeedham, MA

Head of Ground Station Development — Software EngineerSep. 2025 – Present

– Leading software development for an autonomous fixed-wing aircraft competing in the AUVSI SUAS competition, targeting precision water delivery to GPS-specified targets using image recognition.

– Architected ground station software integrating ArduPilot, MissionPlanner, and ROS/MAVROS systems to bridge an on-board NVIDIA Jetson computer with ground controls.

– Engineered a robust fly-by-wire telemetry pipeline, achieving a consistent sub-100ms round-trip latency for real-time sensor monitoring and safety interlocks.

Geeky InsightsRemote

Full-Stack Android App Developer — InternshipApr. 2024 – Apr. 2025

– Designed end-to-end UI/UX of a customizable audio alarm clock app using Figma and Adobe Creative Suite, prioritizing accessibility and intuitive design for a diverse user base.

– Engineered an end-to-end frontend/backend architecture using Flutter, Dart, and Java, optimizing the compilation pipeline into a lightweight, bug-free 15 MB production package.

– Optimized application memory layout to support low-resource devices running Android 9+, reducing RAM consumption to match the baseline footprint of Android’s native system clock.

ACTIVITIES

Olin ICPCNeedham, MA

Competitive Programmer — Team MemberJan. 2026 – Present

– Solving high-difficulty algorithmic problems under competitive time constraints in bi-weekly ICPC-style contests.

– Applying advanced techniques including segment/Fenwick trees, dynamic programming, graph algorithms, string algorithms, network flow, and computational geometry.